Migon Choi

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Professional Summary

AI Scientist focused on decision-centric machine learning for scientific discovery, with dual expertise in laboratory experimentation and data-driven modeling. Specializing in uncertainty-aware modeling and data-efficient workflows, I design the "reasoning layer" of AI-driven experimental systems - integrating experiment, modeling, and adaptive decision-making under incomplete and heterogeneous data regimes. My work positions machine learning as the strategic engine guiding autonomous and scalable scientific platforms.

Education

Duke University - Ph.D., in Materials Science and Engineering Sep 2021 – Present, Durham, NC, USA

- Advised by Catherine Brinson, Expected graduation: May 2026
- NSF National Research Traineeship (NRT) Fellow in AI for Understanding and Designing Materials (aiM)
- Research Focus: Data-efficient machine learning & experiment-integrated materials discovery
- Selected Media & Talks
    - "The Future of Materials Science Starts Here with AI" – Duke University (Study International)
    - "AI + Materials at Duke Pratt" – Duke Pratt News
    - "AI-Guided Materials Discovery" – Duke aiM Program (Technical Talk)

Pusan National University - B.S. in Materials Science and Engineering Mar 2017- Aug 2020, Busan, Korea

- GPA 4.22/4.5 (Top 5%, Summa Cum Laude)

Selected Research: AI for Scientific Discovery under Data Scarcity & Uncertainty

Choi, M et al. *"Uncertainty-Aware Dimensionality Prediction of Low-Dimensional Perovskites by Integrating Bayesian Modeling and Experiments"* (**Manuscript under peer review**)

- **Problem:** Predicting perovskite dimensionality from small, noisy datasets.
- **Approach:** Designed a Bayesian Additive Regression Trees (BART) framework with posterior uncertainty quantification; engineered molecular descriptors from validated SMILES datasets using RDKit.
- **Outcome:** Enabled risk-calibrated material selection by modeling posterior predictive distributions rather than point estimates.

Choi, M et al. *"Chemical Information-Driven Machine Learning Models for Enhanced Performance Prediction of 2D Perovskite Solar Cells"*, ***Solar Energy***, (2026), https://doi.org/10.1016/j.solener.2025.114112

- **Problem:** Predicting perovskite solar cell performance from small, biased experimental datasets.
- **Approach:** Developed an interpretable ML framework to decode the complex relationship between chemical descriptors of spacer cations and device performance.
- **Outcome:** Enabled data-driven identification, narrowing the experimental search space for next-generation solar cell materials.

Choi, M et al. *"P-type Charge Transfer Doping for Mixed Tin-Lead Halide Perovskites: Mo(tfd-COCF$_3$)$_3$ and F$_4$TCNQ"*, ***ACS Applied Materials and Interfaces***, (2025), https://doi.org/10.1021/acsami.5c19800

- **Problem:** Comparing two dopants for p-type doping in mixed Sn-Pb perovskites.
- **Approach:** Led experimental synthesis and doping characterization to understand doping effect in Sn/Pb perovskites.
- **Outcome:** Demonstrated electronic impact of doping on thin film, establishing high-fidelity ground truth data necessary for predictive modeling.

Experience

Engineer - Samsung Electronics Dec 2020 - July 2021, Hwaseong, Korea

- Designed and analyzed 8nm and 14nm RF FinFET materials architectures, improving device performance by mitigating body effects through electrical property and process-structure analysis.
- Collaborated with cross-functional fabrication teams (photolithography, CVD, process integration) to support timely FinFET fabrication for fabless customers.

Senior Associate - Duke Capital Partners (DCP) Oct 2024 – Present, Durham, NC, USA

- Evaluated deep-tech startups in AI, materials, and infrastructure domains, assessing technical feasibility, scalability constraints, and uncertainty risks in early-stage technologies.

Consultant, Duke Advanced Consulting Club (DACC) Sep 2024 – March 2025, Durham

- Led Customer segmentation analysis for a Canada-based wellness franchise, using data-driven insights to identify growth opportunities and optimize marketing strategies.
- Partnerships Lead on the 2025 Southeast Case Competition Steering Committee; collaborated with consultants to organize logistics for 50+ participants.

Mentor - Research Triangle REU Summers 2023 & 2024, Durham, NC, USA

- Led undergraduate researchers on projects involving molecular doping and hybrid perovskite structures.
- Guided students in experimental design, data analysis, and preparing poster presentations for research symposia.

Skills and Tools

Machine Learning & AI Systems: Bayesian modeling (BART); Uncertainty quantification; Active learning & adaptive experiment design; Small-data regime modeling; ML pipeline construction from raw chemical representations (SMILES, RDKit); Model calibration and reliability analysis

Programming: Python (RDKit, ML workflows); Scientific data processing pipelines

Experimental Integration: Thin-film synthesis & characterization; Hall-effect transport measurements; XPS, XRD, SEM; Experiment-in-the-loop ML validation

Presentations & Posters

- **Fpi-16 Conference** - Poster Presentation (July 2025); "*P-type Charge-Transfer Doping in Mixed Sn-Pb Perovskites*"
- **Triangle Student Research Competition** - Poster Presentation (Sep 2025); "*P-type Charge-Transfer Doping in Mixed Sn-Pb Perovskites*"
- **Materials Research Data Alliance** - Poster Presentation (Feb 2023); "*Chemical Information-Driven Machine Learning Models for Enhanced Performance Prediction of 2D Perovskite Solar Cells*"

Hackathons & Data Competitions

- **Schonfeld Hege Fund Datathon** (April 2024), "*Feature engineering for equity price signals*"
- **Joint-NRT Hackathon** (January 2023), "*DFT-bypass ML workflow for predicting exciton binding energies in perovskite solar cells*"

Leadership & Professional Service

AI for Understanding and Designing Materials (aiM) program Student Council Fall 2022 – Spring 2023

- Led peer-to-peer education sessions for graduate trainees
- Hosted social and community-building events

"Ph.D. to CEO" Seminar Series (Duke-funded) Spring & Fall 2025

- Designed and hosted a carrer-transition seminar series connecting Ph.D. students with startup founders, investors, and industry leaders

Materials Research Society (MRS) at Duke Fall 2023 – Spring 2024

- Supported seminar and outreach activities, including science-education events at local libraries